

CS 331 – Database Systems

Assignment #3 – Functional Dependencies

1. (20%) What update anomalies could occur in the following relation:

BOOK_PUBLISHER

<u>Book_id</u>	Title	Publisher_Name	Publisher_Address	Publisher_Phone
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2. (30%) Consider a relation $R(A,B,C,D,E,F,G,H,I,J)$ with the following functional dependencies:

$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

$H \rightarrow J$

- Find all candidate keys of R.
 - Normalize R into 2NF.
 - Normalize R into 3NF.
3. (20%) Consider the following relation:

CAR_SALE

<u>Car_id</u>	Date_sold	<u>Salesman_id</u>	Commision%	Discount_amount
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Assume that a car may be sold by multiple salesmen and hence $\{Car_id, Salesman_id\}$ is the primary key. Additional functional dependencies are:

$Date_sold \rightarrow Discount_amount$

$Salesman_id \rightarrow Commission\%$

$Car_id \rightarrow Date_sold$

- Normalize CAR_SALE into 2NF
 - Normalize CAR_SALE into 3NF
4. (30%) Consider the following relation:

MEDICAL

<u>Physician_id</u>	<u>Patient_id</u>	<u>Date</u>	Diagnosis	Treatment_Code	Charge
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A tuple describes a visit of a patient to a doctor along with a treatment code and charge. Assume that a diagnosis is determined (uniquely) for each patient by a doctor and that each treatment code has a fixed charge (regardless of the patient).

- Infer a set of (two) functional dependencies for MEDICAL from this description.
- Normalize MEDICAL into 2NF
- Normalize MEDICAL into 3NF